

Appendix A. Additional Prior Work

Expression-based DRP requires feature selection to improve interpretability and reduce overfitting, in addition to supervised prediction via classification or survival analysis. *Wrapper* and *filter* algorithms use a predictive model or model-agnostic measure to score features, respectively; for example, AtheroSpectrum used wrapper-style search over random gene subsets to construct the CR-30 signature and then integrated this signature with clinical covariates¹. Wrapper methods are often better aligned with predictive objectives but are computationally expensive and can be unstable in high-dimensional regimes, whereas filter methods are computationally efficient but can be weakly coupled to the final prediction task. Embedded methods, such as L1-penalized logistic regression, optimize prediction and feature sparsity jointly and are therefore widely used in transcriptomic prediction pipelines. Because of their interpretability, generalized linear models are widely adopted to estimate disease risk^{2–4}; for instance, the coefficients of a logistic regression provide an interpretation of how gene expression changes affect the disease risk log-odds. Machine learning models like random forests or boosting classifiers have also been used for DRP^{5,6} and allow for learning interactions between features⁷, though these approaches are generally considered less interpretable than linear models and thus potentially less appropriate for clinical use⁸. Further, since machine learning models typically require more samples than linear regressors and most prior expression-based DRP studies have small sample sizes, gene expression is often imputed from genetic or epigenetic data⁹; however, gene expression is difficult to predict from genomics data¹⁰, with a small proportion of genes being able to be predicted with $r^2 > 0.1$ ^{11,12}. Across these approaches, the simplifying assumptions of additive linear effects^{4,13} or unstructured interactions among gene expression features^{5,6} poorly align with the coordinated gene activity driving disease-relevant cellular processes.

Appendix B. Variational Inference

The Poisson factorization, spike-and-slab structure, and supervised logistic term produce a coupled posterior over all latent variables, so exact inference is intractable. We use mean-field variational inference with coordinate-ascent updates, rendering the supervised term conditionally quadratic by Pólya–Gamma (PG) data augmentation¹⁴.

Appendix B.1. Pólya–Gamma Augmentation of the Supervised Term.

The only non-conjugate component of the model is the Bernoulli supervision

$$p(y_{jk} \mid \theta_j, v_k, \gamma_k, \mathbf{x}_j^{\text{aux}}) = \sigma(A_{jk})^{y_{jk}} (1 - \sigma(A_{jk}))^{1-y_{jk}}, \quad A_{jk} = \theta_j v_k^\top + \mathbf{x}_j^{\text{aux}} \gamma_k^\top,$$

where σ is the logistic function and $\theta_j, v_k, \mathbf{x}_j^{\text{aux}}, \gamma_k$ are row vectors. Following Polson, Scott, and Windle¹⁴, for a Pólya–Gamma variable $\zeta \sim \text{PG}(1, 0)$,

$$\frac{(e^A)^y}{1 + e^A} = \frac{1}{2} e^{\tilde{\kappa} A} \int_0^\infty e^{-\zeta A^2/2} p_{\text{PG}}(\zeta \mid 1, 0) d\zeta, \quad \tilde{\kappa} = y - \frac{1}{2}. \quad (\text{B.1})$$

Introducing one auxiliary variable ζ_{jk} per sample–phenotype pair gives the augmented likeli-

hood

$$p(y_{jk}, \zeta_{jk} \mid A_{jk}) \propto \exp\left(\tilde{\kappa}_{jk} A_{jk} - \frac{\zeta_{jk}}{2} A_{jk}^2\right) p_{\text{PG}}(\zeta_{jk} \mid 1, 0), \quad \tilde{\kappa}_{jk} = y_{jk} - \frac{1}{2}. \quad (\text{B.2})$$

We write the PG shift as $\tilde{\kappa}_{jk}$ to distinguish it from the number of phenotypes κ . Equation (B.1) is an exact identity rather than a variational bound, so the augmented model has the same marginal likelihood as the original. Conditional on ζ_{jk} the exponent is quadratic in A_{jk} , which is conjugate to the Gaussian factors $q(v_k)$ and $q(\gamma_k)$ but only log-quadratic in θ_j ; the consequences for the θ update are given in the remark below.

Appendix B.2. Joint Distribution and Variational Family.

Conditioned on the auxiliary covariates and including the augmentation variables, the joint distribution factorizes as

$$\begin{aligned} p(\mathbf{X}, \mathbf{Y}, \boldsymbol{\theta}, \boldsymbol{\xi}, \tilde{\boldsymbol{\beta}}, M, \pi, \eta, \mathbf{v}, s, \boldsymbol{\gamma}, \boldsymbol{\zeta} \mid \mathbf{X}^{\text{aux}}) &= \prod_{g=1}^p \left[p(\pi_g) p(\eta_g) \prod_{\ell=1}^d p(m_{g\ell} \mid \pi_g) p(\tilde{\beta}_{g\ell} \mid \eta_g) \right] \\ &\times \prod_{j=1}^n \left[p(\xi_j) \prod_{\ell=1}^d p(\theta_{j\ell} \mid \xi_j) \prod_{g=1}^p p(x_{jg} \mid \theta_j, \beta_g) \right] \\ &\times \prod_{k=1}^{\kappa} \left[p(\gamma_k) \prod_{\ell=1}^d p(s_{k\ell}) p(v_{k\ell} \mid s_{k\ell}) \prod_{j=1}^n p(y_{jk}, \zeta_{jk} \mid \theta_j, v_k, \gamma_k, \mathbf{x}_j^{\text{aux}}) \right]. \end{aligned}$$

We assume the fully factorized variational family

$$\begin{aligned} q(\eta_g) &= \text{Gamma}(a_{\eta_g}, b_{\eta_g}), & q(\tilde{\beta}_{g\ell}) &= \text{Gamma}(a_{\tilde{\beta}_{g\ell}}, b_{\tilde{\beta}_{g\ell}}), \\ q(m_{g\ell}) &= \text{Bernoulli}(\rho_{g\ell}), & q(\pi_g) &= \text{Beta}(a_{\pi_g}, b_{\pi_g}), \\ q(\xi_j) &= \text{Gamma}(a_{\xi_j}, b_{\xi_j}), & q(\theta_{j\ell}) &= \text{Gamma}(a_{\theta_{j\ell}}, b_{\theta_{j\ell}}), \\ q(v_{k\ell}) &= \mathcal{N}(\mu_{v_{k\ell}}, \tau_{v_{k\ell}}^2), & q(s_{k\ell}^{-1}) &= \text{InvGaussian}(\mu_{s_{k\ell}}, \lambda_{s_{k\ell}}), \\ q(\gamma_k) &= \mathcal{N}(\mu_{\gamma_k}, \Sigma_{\gamma_k}), & q(\zeta_{jk}) &= \text{PG}(1, c_{jk}), \end{aligned}$$

where the Bayesian-Lasso factor is placed on the *precision* $s_{k\ell}^{-1}$, which is inverse-Gaussian under the Gaussian-exponential scale mixture. For any Gamma factor $q(X) = \text{Gamma}(a, b)$ we use $\mathbb{E}_q[X] = a/b$, $\mathbb{E}_q[\log X] = \Psi(a) - \log b$, and $\text{Var}_q(X) = a/b^2$, where $\Psi(\cdot)$ is the digamma function.

Optimal augmentation factor. Because the augmented exponent in (B.2) is $-\frac{1}{2}\zeta_{jk}A_{jk}^2$, the coordinate-optimal factor for ζ_{jk} is again Pólya-Gamma, exponentially tilted with $c_{jk} = \sqrt{\mathbb{E}_q[A_{jk}^2]}$. The only moment used downstream is its mean,

$$\bar{\zeta}_{jk} \equiv \mathbb{E}_q[\zeta_{jk}] = \frac{1}{2c_{jk}} \tanh\left(\frac{c_{jk}}{2}\right), \quad \lim_{c_{jk} \rightarrow 0} \bar{\zeta}_{jk} = \frac{1}{4}, \quad (\text{B.3})$$

evaluated by a second-order Taylor expansion of $\tanh$ when c_{jk} is small.

Count allocation and gene-side updates. Following Raikov's decomposition theorem¹⁵, we introduce latent allocations $z_{jg\ell}$ with $x_{jg} = \sum_{\ell=1}^d z_{jg\ell}$ and $z_{jg\ell} \sim \text{Poisson}(\theta_{j\ell}\beta_{g\ell})$. Because

$\beta_{g\ell} = m_{g\ell}\tilde{\beta}_{g\ell}$, inactive gene-program pairs contribute zero rate automatically. Writing $e_{jg\ell} = \exp(\Psi(a_{\theta_{j\ell}}) - \log b_{\theta_{j\ell}} + \Psi(a_{\tilde{\beta}_{g\ell}}) - \log b_{\tilde{\beta}_{g\ell}})$, the optimal multinomial allocation is

$$\phi_{jg\ell} = \frac{\rho_{g\ell} e_{jg\ell}}{\sum_{\ell'=1}^d \rho_{g\ell'} e_{jg\ell'}}, \quad \mathbb{E}_q[z_{jg\ell}] = x_{jg}\phi_{jg\ell}, \quad (\text{B.4})$$

and the slab loading update is

$$a_{\tilde{\beta}_{g\ell}} = \alpha_\beta + \sum_{j=1}^n x_{jg}\phi_{jg\ell}, \quad b_{\tilde{\beta}_{g\ell}} = \mathbb{E}_q[\eta_g] + \rho_{g\ell} \sum_{j=1}^n \mathbb{E}_q[\theta_{j\ell}]. \quad (\text{B.5})$$

For the Bernoulli membership variable, the log-odds update compares the expected log-joint under $m_{g\ell} = 1$ and $m_{g\ell} = 0$. Let $\bar{\phi}_{jg\ell} = e_{jg\ell}(e_{jg\ell} + \sum_{m \neq \ell} \rho_{gm} e_{jgm})^{-1}$ be the allocation with program ℓ held active, and define

$$L_{jg\ell} = x_{jg} \bar{\phi}_{jg\ell} \left(\Psi(a_{\theta_{j\ell}}) - \log b_{\theta_{j\ell}} + \Psi(a_{\tilde{\beta}_{g\ell}}) - \log b_{\tilde{\beta}_{g\ell}} \right) - \mathbb{E}_q[\theta_{j\ell}] \mathbb{E}_q[\tilde{\beta}_{g\ell}] - \log \Gamma(x_{jg} \bar{\phi}_{jg\ell} + 1);$$

the final term evaluates $\mathbb{E}_q[\log z_{jg\ell}!]$ at $\mathbb{E}_q[z_{jg\ell}]$ rather than exactly. The update is

$$\log \frac{\rho_{g\ell}}{1 - \rho_{g\ell}} = \Psi(a_{\pi_g}) - \Psi(b_{\pi_g}) + \sum_{j=1}^n L_{jg\ell}. \quad (\text{B.6})$$

The conjugate hyperparameter updates are

$$a_{\pi_g} = \alpha_\pi + \sum_{\ell=1}^d \rho_{g\ell}, \quad b_{\pi_g} = \beta_\pi + d - \sum_{\ell=1}^d \rho_{g\ell}, \quad (\text{B.7})$$

$$a_{\eta_g} = \alpha_\eta + d\alpha_\beta, \quad b_{\eta_g} = \lambda_\eta + \sum_{\ell=1}^d \frac{a_{\tilde{\beta}_{g\ell}}}{b_{\tilde{\beta}_{g\ell}}}, \quad a_{\xi_j} = \alpha_\xi + d\alpha_\theta, \quad b_{\xi_j} = \lambda_\xi + \sum_{\ell=1}^d \frac{a_{\theta_{j\ell}}}{b_{\theta_{j\ell}}}. \quad (\text{B.8})$$

Pathway mask. In the de novo setting all gene-program pairs are updated freely. In the pathway-guided setting we fix $m_{g\ell} = 0$, $\rho_{g\ell} = 0$, and $\beta_{g\ell} = 0$ for every entry with $M_{g\ell} = 0$, which implies $\phi_{jg\ell} = 0$ for all j ; the same updates are applied to the allowed entries only. The gene-side sums over ℓ in (B.7)–(B.8) run over allowed entries, so d is replaced by $d_g = \sum_\ell M_{g\ell}$ in those three updates. The sample-side count d in a_{ξ_j} is unchanged, since θ is not masked.

Supervised updates under augmentation. Conditional on $q(\zeta)$, the augmented supervised term (B.2) is quadratic in A_{jk} , and $\bar{\zeta}_{jk}$ plays the role of the quadratic-curvature coefficient. Throughout we use the leave-one-out predictor $C_{jk}^{(-\ell)} = \sum_{m \neq \ell} \mathbb{E}_q[\theta_{jm}] \mathbb{E}_q[v_{km}] + \mathbf{x}_j^{\text{aux}} \mu_{\gamma_k}$, i.e. A_{jk} with the ℓ -th program term removed. The sample-activity update is

$$a_{\theta_{j\ell}} = \alpha_\theta + \sum_{g=1}^p x_{jg}\phi_{jg\ell}, \quad b_{\theta_{j\ell}} = \mathbb{E}_q[\xi_j] + \sum_{g=1}^p \rho_{g\ell} \mathbb{E}_q[\tilde{\beta}_{g\ell}] + R_{j\ell}, \quad (\text{B.9})$$

$$R_{j\ell} = \sum_{k=1}^{\kappa} \left[-\bar{\kappa}_{jk} \mathbb{E}_q[v_{k\ell}] + \bar{\zeta}_{jk} \mathbb{E}_q[v_{k\ell}] C_{jk}^{(-\ell)} + \frac{1}{2} \bar{\zeta}_{jk} \mathbb{E}_q[v_{k\ell}^2] \mathbb{E}_q[\theta_{j\ell}] \right]. \quad (\text{B.10})$$

The supervised correction $R_{j\ell}$ enters the Gamma rate of $\theta_{j\ell}$ and couples program activity to disease status: the label-linear term shifts disease-relevant programs ($v_{k\ell} \neq 0$) toward higher

activity in cases and lower activity in controls, while the $\bar{\zeta}_{jk}$ -scaled terms damp that shift by the curvature of the logistic.

Remark (exactness of the θ update). Conditional on ζ , the terms in $\theta_{j\ell}$ contributed by (B.2) are $\tilde{\kappa}_{jk} v_{k\ell} \theta_{j\ell} - \bar{\zeta}_{jk} v_{k\ell} C_{jk}^{(-\ell)} \theta_{j\ell} - \frac{1}{2} \bar{\zeta}_{jk} v_{k\ell}^2 \theta_{j\ell}^2$, so the coordinate-optimal factor is $q^*(\theta_{j\ell}) \propto \theta_{j\ell}^{a-1} \exp(-b\theta_{j\ell} - c\theta_{j\ell}^2)$ with $c = \frac{1}{2} \sum_k \bar{\zeta}_{jk} \mathbb{E}_q[v_{k\ell}^2]$. The quadratic term lies outside the span of the Gamma sufficient statistics $(\log \theta, \theta)$, so q^* is not Gamma. Equation (B.9) therefore restricts $q(\theta_{j\ell})$ to the Gamma family and linearizes $c\theta_{j\ell}^2$ about the current mean, replacing it with $(c\mathbb{E}_q[\theta_{j\ell}])\theta_{j\ell}$; the resulting update is a fixed-point iteration rather than an exact coordinate maximizer. Because the linearized rate may be negative, it is floored at a small $\varepsilon > 0$, $b_{\theta_{j\ell}} \leftarrow \max(b_{\theta_{j\ell}}, \varepsilon)$, which is a projection back into the family.

Regression and scale updates. The regression coefficient updates are

$$\tau_{v_{k\ell}}^{-2} = \mathbb{E}_q[s_{k\ell}^{-1}] + \sum_{j=1}^n \bar{\zeta}_{jk} \mathbb{E}_q[\theta_{j\ell}^2], \quad \mu_{v_{k\ell}} = \tau_{v_{k\ell}}^2 \sum_{j=1}^n [\tilde{\kappa}_{jk} - \bar{\zeta}_{jk} C_{jk}^{(-\ell)}] \mathbb{E}_q[\theta_{j\ell}], \quad (\text{B.11})$$

with $\mathbb{E}_q[\theta_{j\ell}^2] = \mathbb{E}_q[\theta_{j\ell}]^2 + \text{Var}_q(\theta_{j\ell})$, and

$$\Sigma_{\gamma_k}^{-1} = \frac{1}{\sigma_{\gamma}^2} I + \sum_{j=1}^n \bar{\zeta}_{jk} (\mathbf{x}_j^{\text{aux}})^{\top} \mathbf{x}_j^{\text{aux}}, \quad \mu_{\gamma_k} = \Sigma_{\gamma_k} \sum_{j=1}^n (\mathbf{x}_j^{\text{aux}})^{\top} [\tilde{\kappa}_{jk} - \bar{\zeta}_{jk} \mathbb{E}_q[\theta_j] \mathbb{E}_q[v_k]^{\top}]. \quad (\text{B.12})$$

The Bayesian-Lasso scale update is unchanged by the augmentation, since the prior on v is untouched. Under $v_{k\ell} | s_{k\ell} \sim \mathcal{N}(0, s_{k\ell})$, $s_{k\ell} \sim \text{Exp}(1/2b_v^2)$, the optimal factor for $s_{k\ell}^{-1}$ is inverse-Gaussian with $\lambda_{s_{k\ell}} = b_v^{-2}$ and

$$\mathbb{E}_q[s_{k\ell}^{-1}] = \mu_{s_{k\ell}} = \frac{1}{b_v \sqrt{\mathbb{E}_q[v_{k\ell}^2]}}. \quad (\text{B.13})$$

Finally, the tilt is refreshed by setting $c_{jk}^2 = \mathbb{E}_q[A_{jk}^2]$. Because the variational factors are independent, $\text{Var}_q(A_{jk})$ decomposes into separate θ -, v -, and γ -contributions,

$$\begin{aligned} \mathbb{E}_q[A_{jk}] &= \mathbb{E}_q[\theta_j] \mu_{v_k}^{\top} + \mathbf{x}_j^{\text{aux}} \mu_{\gamma_k}, \\ c_{jk}^2 &= \mathbb{E}_q[A_{jk}^2] = \mathbb{E}_q[A_{jk}]^2 + \sum_{\ell=1}^d [\text{Var}_q(\theta_{j\ell}) \mathbb{E}_q[v_{k\ell}^2] + \mathbb{E}_q[\theta_{j\ell}]^2 \tau_{v_{k\ell}}^2] + \mathbf{x}_j^{\text{aux}} \Sigma_{\gamma_k} (\mathbf{x}_j^{\text{aux}})^{\top}, \end{aligned} \quad (\text{B.14})$$

with $\mathbb{E}_q[v_{k\ell}^2] = \mu_{v_{k\ell}}^2 + \tau_{v_{k\ell}}^2$, after which $\bar{\zeta}_{jk}$ is recomputed via (B.3). All three variance terms are required for the tilt to equal the full second moment, so that the residual in (B.15) vanishes at the optimum.

Objective and optimization. The variational objective is the evidence lower bound of the augmented model, $\mathcal{L}(q) = \mathbb{E}_q[\log p(\mathbf{X}, \mathbf{Y}, \boldsymbol{\theta}, \boldsymbol{\xi}, \tilde{\boldsymbol{\beta}}, \mathbf{M}, \boldsymbol{\pi}, \boldsymbol{\eta}, \mathbf{v}, \mathbf{s}, \boldsymbol{\gamma}, \boldsymbol{\zeta} | \mathbf{X}^{\text{aux}})] - \mathbb{E}_q[\log q]$, combining the expected log-priors, the augmented Poisson likelihood, the augmented supervised term, and the entropies of all factors. Marginalizing the optimal PG factor analytically, the supervised contribution at $q(\zeta_{jk}) = \text{PG}(1, c_{jk})$ reduces to

$$\mathcal{L}^{\text{sup}} = \sum_{j=1}^n \sum_{k=1}^{\kappa} \left[\tilde{\kappa}_{jk} \mathbb{E}_q[A_{jk}] - \frac{c_{jk}}{2} + \log \sigma(c_{jk}) - \frac{\bar{\zeta}_{jk}}{2} (\mathbb{E}_q[A_{jk}^2] - c_{jk}^2) \right], \quad (\text{B.15})$$

whose last term vanishes when the tilt is at its optimum $c_{jk}^2 = \mathbb{E}_q[A_{jk}^2]$. Because (B.1) is exact rather than a bound, $\mathcal{L}(q)$ is a valid lower bound on $\log p(\mathbf{X}, \mathbf{Y} \mid \mathbf{X}^{\text{aux}})$ at every iteration. Monotonicity holds for every block whose update is an exact coordinate maximizer, but not for θ , whose update is linearized; we therefore monitor the ELBO empirically and base model selection and early stopping on validation performance, with all reported predictive results evaluated on held-out data. Algorithm 2 summarizes the procedure.

Complexity. Let $\text{nnz}(\mathbf{X}) = |\{(j, g) : x_{jg} \neq 0\}|$. The allocation (B.4) is evaluated on the fly for each nonzero and never materialized, giving $O(\text{nnz}(\mathbf{X})d)$ time and $O(d)$ working memory per entry; the loading update (B.5) reuses the same sums. In (B.6) the term $\mathbb{E}_q[\theta_{j\ell}]\mathbb{E}_q[\tilde{\beta}_{g\ell}]$ is nonzero when $x_{jg} = 0$, but factorizes as $(\sum_j \mathbb{E}_q[\theta_{j\ell}])\mathbb{E}_q[\tilde{\beta}_{g\ell}]$, so its cost is $O(pd)$ rather than $O(npd)$. Accumulating $\sum_m \mathbb{E}_q[\theta_{jm}]\mathbb{E}_q[v_{km}]$ once per (j, k) and subtracting the ℓ -th term makes all $C_{jk}^{(-\ell)}$ available in $O(nd\kappa)$ rather than $O(nd^2\kappa)$, which also bounds (B.11) and (B.14); the auxiliary update (B.12) costs $O(n(p^{\text{aux}})^2 + (p^{\text{aux}})^3)$ per phenotype. A training sweep is therefore $O(\text{nnz}(\mathbf{X})d + nd\kappa + \kappa\{n(p^{\text{aux}})^2 + (p^{\text{aux}})^3\})$, dominated by the first term whenever $p^{\text{aux}} \ll p$. Memory is $O(\text{nnz}(\mathbf{X}) + (n + p)d + \kappa(d + (p^{\text{aux}})^2))$. Fold-in for one held-out sample costs $O(\text{nnz}(\mathbf{x}_{j^*})d)$ per local iteration and touches no global parameter, so test-time inference is embarrassingly parallel across samples.

Test- and validation-time inference (fold-in). At test time the labels y_{j^*k} are unobserved, so (B.2) contributes no likelihood factor for a new sample j^* . We freeze the *global* parameters—the gene-side factors $\{a_{\tilde{\beta}}, b_{\tilde{\beta}}, \rho, a_{\eta}, b_{\eta}, a_{\pi}, b_{\pi}\}$ and the regression coefficients $\{\mu_v, \tau_v^2, \mu_{\gamma}, \Sigma_{\gamma}\}$ —and infer only the *local* variables $\{\theta_{j^*\ell}, \xi_{j^*}\}$ from the counts x_{j^*g} through the allocations $\phi_{j^*g\ell}$. The supervised correction vanishes, $R_{j^*\ell} = 0$, no PG variables ζ_{j^*k} are needed, and (B.9) reduces to its unsupervised form

$$a_{\theta_{j^*\ell}} = \alpha_{\theta} + \sum_{g=1}^p x_{j^*g} \phi_{j^*g\ell}, \quad b_{\theta_{j^*\ell}} = \mathbb{E}_q[\xi_{j^*}] + \sum_{g=1}^p \rho_{g\ell} \mathbb{E}_q[\tilde{\beta}_{g\ell}]. \quad (\text{B.16})$$

Note that θ_{j^*} is thus driven only by held-out expression and the fixed gene programs, whereas training activities are label-informed through $R_{j\ell}$. Given the converged fold-in posterior, the predictor moments are read off from (B.14) as $\mathbb{E}_q[A_{j^*k}] = \mathbb{E}_q[\theta_{j^*}]\mu_{v_k}^{\top} + \mathbf{x}_{j^*}^{\text{aux}}\mu_{\gamma_k}$ and $\text{Var}_q(A_{j^*k}) = c_{j^*k}^2 - \mathbb{E}_q[A_{j^*k}]^2$, and the posterior predictive probability is approximated by the Gaussian–logistic integral

$$\hat{y}_{j^*k} = \mathbb{E}_q[\sigma(A_{j^*k})] \approx \sigma\left(\frac{\mathbb{E}_q[A_{j^*k}]}{\sqrt{1 + \pi \text{Var}_q(A_{j^*k})/8}}\right), \quad (\text{B.17})$$

where π denotes the mathematical constant. This reduces to the plug-in estimate $\sigma(\mathbb{E}_q[A_{j^*k}])$ in the zero-variance limit. The same routine supplies the validation predictions used for model selection in Algorithm 2, and is summarized in Algorithm 3.

Algorithm 2 Pólya–Gamma CAVI for DRGP

Input: Expression $\mathbf{X}$, labels $\mathbf{Y}$, auxiliary covariates $\mathbf{X}^{\text{aux}}$, number of factors d , pathway mask $\mathbf{M}$, and model hyperparameters

Output: Fitted global and local variational parameters

- 1: Initialize all variational parameters and enforce $\mathbf{M}$
 - 2: **while** the ELBO has not converged **do**
 - 3: Update count allocations ϕ ▷ Eq. (B.4)
 - 4: Update gene loadings and memberships $(\tilde{\beta}, \rho)$ ▷ Eqs. (B.5), (B.6)
 - 5: Update gene-level parameters (π, η) ▷ Eq. (B.7), (B.8)
 - 6: Update sample activities (θ, ξ) ▷ Eqs. (B.9), (B.8)
 - 7: Update supervised parameters (s, v, γ) ▷ Eqs. (B.13)–(B.12)
 - 8: Update Pólya–Gamma moments c and $\bar{\zeta}$ ▷ Eqs. (B.14), (B.3)
 - 9: **end while**
 - 10: **return** fitted variational parameters
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Algorithm 3 Fold-in inference and risk prediction

Input: Held-out expression $\mathbf{x}_{j^*}$, auxiliary covariates $\mathbf{x}_{j^*}^{\text{aux}}$, and frozen global parameters from Algorithm 2

Output: Predicted risks $\hat{y}_{j^*k}$

- 1: **for** each held-out sample j^* **do**
 - 2: Initialize local parameters $(\phi_{j^*}, \theta_{j^*}, \xi_{j^*})$
 - 3: **while** the local Poisson objective has not converged **do**
 - 4: Update ϕ_{j^*} , then θ_{j^*} with $R_{j^*\ell} = 0$, then ξ_{j^*} ▷ Eqs. (B.4), (B.16), (B.8)
 - 5: **end while**
 - 6: **for** each phenotype k **do**
 - 7: Compute posterior moments of A_{j^*k} and predict $\hat{y}_{j^*k}$ ▷ Eqs. (B.14), (B.17)
 - 8: **end for**
 - 9: **end for**
 - 10: **return** $\{\hat{y}_{j^*k}\}$
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Appendix C. Ranking Genes within DRGP Programs

For all downstream analyses requiring a continuous ranking of genes within a DRGP program, we use

$$\hat{s}_{g\ell} = \mathbb{E}_q[\beta_{g\ell}] \mathbb{E}_q[\eta_g]. \quad (\text{C.1})$$

The expected loading $\mathbb{E}_q[\beta_{g\ell}]$ incorporates the posterior gene–program inclusion probability, while $\mathbb{E}_q[\eta_g]$ adjusts for gene-specific loading capacity. Consequently, $\hat{s}_{g\ell}$ favors genes that load strongly on program ℓ but do not load broadly across many programs.

For biological program summaries, genes were ordered by $\hat{s}_{g\ell}$ and the top 50 genes were retained where a fixed-size list was required. In simulation, support recovery was evaluated using $\hat{s}_{\cdot\ell}$ as the continuous gene score, with fitted programs Hungarian-matched to simulated programs by signed cosine similarity of the gene-score vectors. Program reproducibility across

random initializations was likewise evaluated by cosine similarity between gene-score vectors.

Appendix D. Hyperparameter selection

For simulations, hyperparameters were fixed before evaluation and held constant across all simulated difficulty settings.

Table D1. Fixed parameters of the simulation framework. Swept parameters (r , w_Z , ρ , δ , φ , w_s) are given in §3.2.

Symbol	Meaning	Single-cell	Bulk
–	individuals / samples	80 (100 cells each)	2,000
p	genes	10,000	5,000
–	cell types	6	–
–	disease programs (annot. + <i>de novo</i>)	2 + 2	2 + 4
–	nuisance programs (annot. + <i>de novo</i>)	0 + 1	1 + 1
$ S_\ell $	program support size	U{50,100}	U{50,150}
α_p, β_p	propensity Beta prior	2.0, 2.0	–
σ_u	per-program propensity jitter	0.5	–
(q_0, q_1)	baseline / max carrier prob.	(0.20, 0.80)	–
λ_ℓ	program prevalence multiplier	1.0 (all ℓ)	–
π_ℓ	nuisance carrier rate	0.5	0.5
ω_ℓ	liability weight per DRGP	{2.0, 0.7, 2.0, 0.7}, balanced signs	$ v = 1.5$
–	nuisance liability weight	0	0
–	liability form / prevalence	probit / 0.5	probit / 0.26
–	activation magnitude	Gamma(2,2)	binary carrier
–	program polarity	up/up/mixed/down/mixed	up
–	truth draws per configuration	5	5

For experimental analyses, hyperparameters were selected separately for each dataset, input representation, cohort size, and prior-knowledge mode using tree-structured Parzen estimator optimization implemented in Optuna¹⁶. The objective was mean validation ROC-AUC. For single-cell analyses, all cells from the same patient were assigned to the same partition to prevent patient-level information leakage between training and validation sets. We evaluated 30 trials for each COVID-19 single-cell configuration and 60 trials for each COVID-19 pseudo-bulk and GTEx configuration.

The search included the number of latent programs K ; the Gamma prior shapes α_θ , α_β , α_ξ , and α_η ; the Laplace scale b_v for the supervised program coefficients; the prior standard deviation σ_γ for auxiliary-covariate coefficients; and the spike-and-slab parameters α_π and $\beta_{\pi, \text{scale}}$. For pathway-masked models, K was fixed to the number of supplied pathways. For combined models, the pathway-constrained factors were fixed and the number of additional *de novo* factors was optimized. The configuration with the highest validation AUC was used for the final fit. Final experimental models were run for at most 500 coordinate-ascent iterations with an ELBO-based relative convergence tolerance of 10^{-3} and a 50-iteration warm-up before updating the supervised coefficients. The selected values for each reported configuration are provided in Table D2.

In contrast, simulation hyperparameters were fixed before evaluation and held constant across all simulated difficulty settings. This design prevents changes in model tuning from being confounded with changes in perturbation strength, responder-cell prevalence, or the transcriptomic and auxiliary liability shares. Both simulation arms used $\alpha_\theta = \alpha_\beta = 0.3$, $\sigma_\gamma = 1$, a convergence tolerance of 10^{-3} , and a maximum of 3,000 coordinate-ascent iterations. The COVID-19 single-cell simulations were fit at each of $d \in \{8, 10, 14\}$ latent programs, with the supervised-coefficient scale held fixed at $b_v = 1$. In the GTEx bulk simulations the number of simulated programs is $d^* = 8$, while the fitted models used $d = 10$ for the unmasked configuration, $d_{\text{path}} = 3$ for the pathway-masked configuration, and 3 pathway-anchored plus 7 de novo factors for the combined configuration; all unsupervised baselines were fit with $d = 10$ for comparability. In this arm b_v was initialized at 1 and auto-calibrated during fitting to the data precision, converging to ≈ 0.01 .

All comparison methods were run with default hyperparameters.

Table D2. **Selected hyperparameters for the experimental configurations.** Values maximize patient- or subject-grouped validation AUC. The number of latent gene programs is written $K_{\text{pathway}} + K_{\text{de novo}}$: unmasked models are entirely *de novo*, masked models are entirely pathway-anchored, and combined models use both.

Dataset/representation	Mode	$K_{\text{pathway}} + K_{\text{de novo}}$	α_θ	α_β	α_ξ	α_η	b_v	σ_γ	α_π	$\beta_{\pi, \text{scale}}$
COVID-19 single-cell	Unmasked	0 + 50	3.0266	2.3343	2.3163	1.7338	0.0639	0.1205	2.7179	1.5597
COVID-19 single-cell	Masked	346 + 0	2.0708	1.0016	2.1426	1.6326	0.2451	1.4332	0.1292	17.2617
COVID-19 single-cell	Combined	346 + 50	2.0708	1.0016	2.1426	1.6326	0.2451	1.4332	0.1292	17.2617
COVID-19 pseudo-bulk	Unmasked	0 + 25	2.3066	0.9487	3.9071	0.8187	0.0932	0.7832	1.8736	8.4993
COVID-19 pseudo-bulk	Masked	351 + 0	2.4135	1.1470	3.8935	3.2130	1.5999	1.4595	1.0371	36.8330
COVID-19 pseudo-bulk	Combined	351 + 50	2.4135	1.1470	3.8935	3.2130	1.5999	1.4595	1.0371	36.8330
GTEx whole blood	Unmasked	0 + 20	3.3541	2.9740	3.0515	3.1994	0.0657	0.2927	0.1573	29.2676
GTEx whole blood	Masked	309 + 0	0.7029	1.4259	0.6661	1.2495	0.8394	1.5575	0.5051	10.4172
GTEx whole blood	Combined	309 + 20	0.7029	1.4259	0.6661	1.2495	0.8394	1.5575	0.5051	10.4172

Appendix E. Additional Results

Appendix E.1. *Simulation Results Across the Parameter Sweep*

The single-cell simulation was run over the difficulty grid of §3.2 rather than a single configuration, summarised here as marginal profiles along each factor. Three patterns hold across the sweeps. Annotated recovery is set by whether a method is given the supports rather than by the difficulty of the instance: DRGP-masked is invariant at 0.801 and DRGP-combined spans 0.762–0.801, while methods without the supports stay near the chance level $|S_\ell|/p = 0.0075$ (Figs. E1, E2). De novo recovery is at chance for every method in every configuration (0.008–0.010). Patient-level prediction responds most to the responder fraction and to the transcriptomic liability share, and least to the perturbation strength: across the sweep the largest swing in AUC is 0.146 for ρ and 0.118 for w_Z , against at most 0.092 for r (Figs. E4, E3).

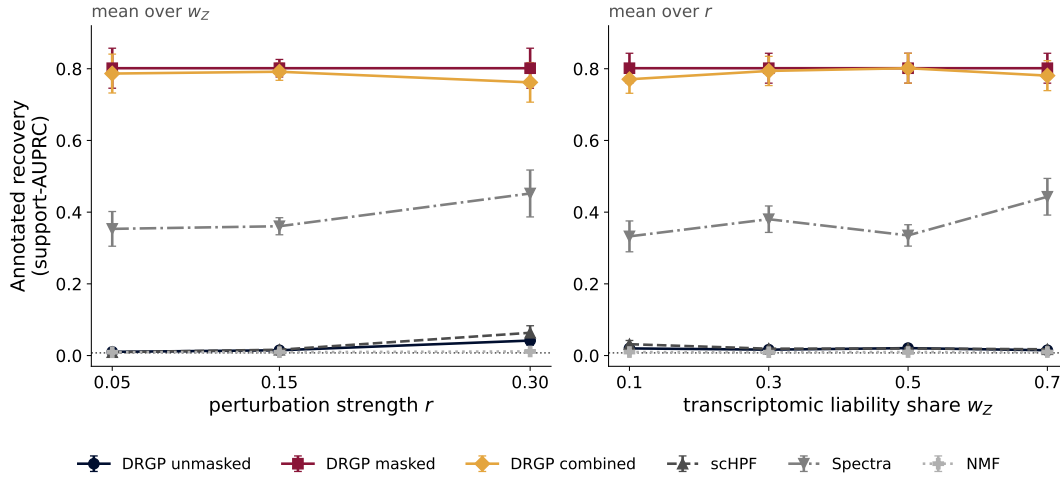


Fig. E1. **Recovery of Pathways (support-AUPRC) as a function of r and w_Z for a fixed $\rho = 0.3$.** Left: Support-AUPRC against r , averaging over $w_Z \in \{0.1, 0.3, 0.5, 0.7\}$. Right: Support-AUPRC against w_Z , averaging over $r \in \{0.05, 0.15, 0.30\}$. Both panels hold $\rho = 0.3$ and $d = 8$. Points are means over the five simulated truth draws with ± 1 standard error, and the dotted line marks chance ($|S_\ell|/p = 0.0075$).

Appendix E.2. *Cell-Level Prediction in the Single-Cell Simulation*

Assigning each cell its individual’s phenotype produces a label that individual cells cannot support. At the headline setting of the simulated grid ($w_Z = 0.7$, $r = 0.30$), cell-level prediction remains at chance for every method (0.50–0.52 AUC, Fig. E5, not because the programs are undetectable, but because a responder fraction of $\rho = 0.3$ leaves most cells in a case individual carrying no DRGP at all. Averaging a patient’s cell-level probabilities recovers the signal that individual cells cannot resolve, making patient-level performance the more appropriate and reliable measure of predictive accuracy in this setting.

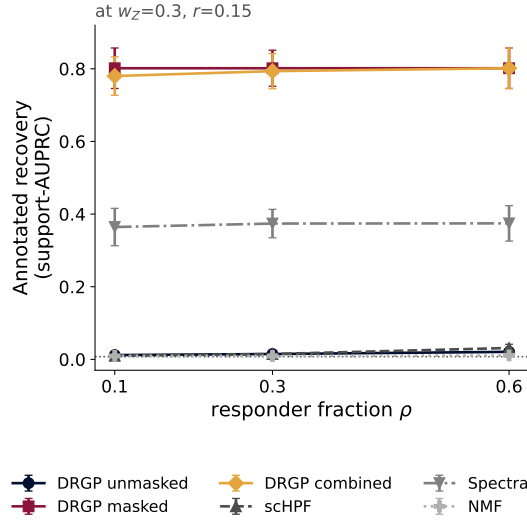


Fig. E2. **Recovery of Pathways (support-AUPRC) on the single-cell simulation as a function of ρ for a fixed $w_Z = 0.3$, $r = 0.15$, $d = 8$.** Points are means over five simulated-truth draws with ± 1 standard error; the dotted line marks chance, the mean support prevalence $|S_\ell|/p = 0.0075$.

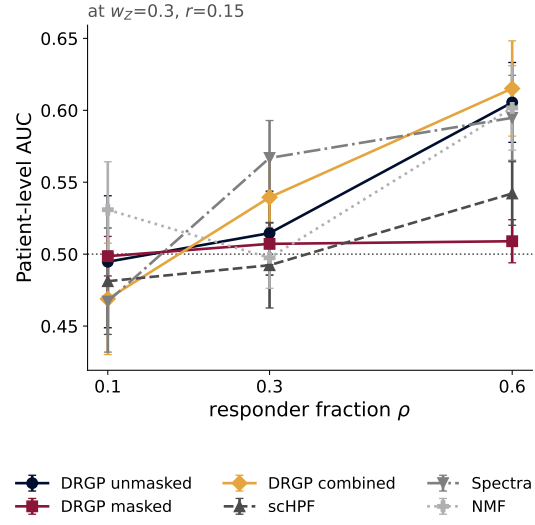


Fig. E3. **Patient-level AUC in the single-cell simulation as a function of ρ for a fixed $w_Z = 0.3$, and $r = 0.15$.** Points are means over five simulated-truth draws with ± 1 standard error; the dotted line marks chance.

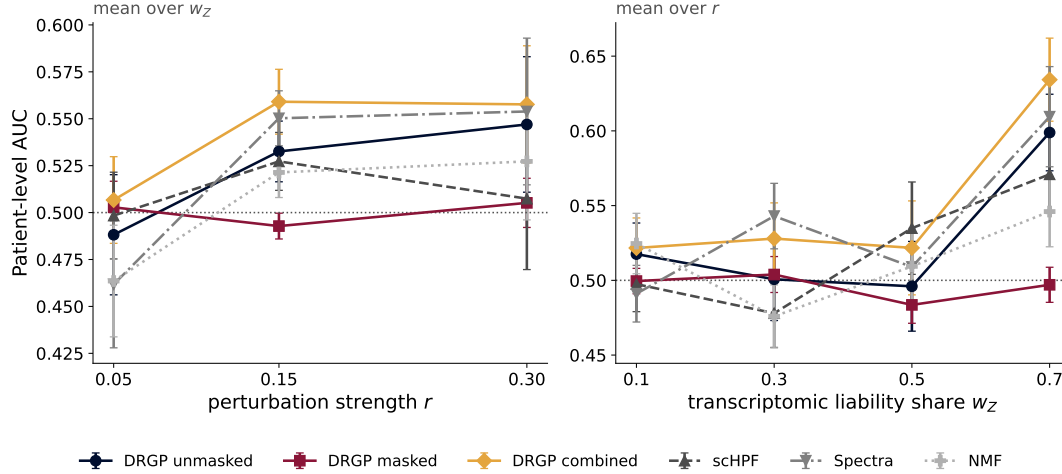


Fig. E4. **Patient-level AUC in the COVID-19 single-cell simulation as a function of r and w_Z for a fixed $\rho = 0.3$.** Left: AUC against r , averaging over $w_Z \in \{0.1, 0.3, 0.5, 0.7\}$. Right: AUC against w_Z , averaging over $r \in \{0.05, 0.15, 0.30\}$. Both panels hold the responder fraction at $\rho = 0.3$ and the latent dimension at $d = 8$. Points are means over the five simulated truth draws per each configuration, and error bars show ± 1 standard error. The dotted line marks chance AUC = 0.5.

Appendix E.3. Reproducible Gene Programs on Experimental Results

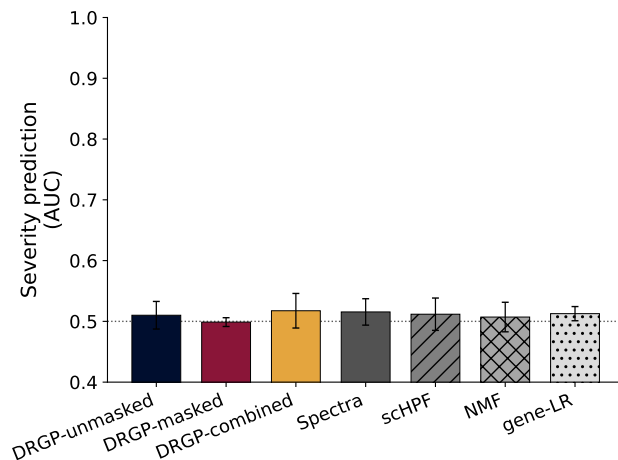


Fig. E5. **All methods remain at chance on simulated DRP.** COVID-19 scRNA-seq simulations evaluated on cells from held-out patients using inherited labels. Bars show means with error bars denoting ± 1 SD. Dotted lines mark chance performance.

Table E1. **Configurations used for program interpretation.** Both cohorts were fit in *combined* mode with five random seeds, varying only the model-initialization seed. d is written $d_{\text{path}} + d_{\text{de novo}}$. Values not listed were left at their defaults; in particular α_{ξ} and α_{η} were not tuned for these runs.

	GTEX whole blood	COVID-19 pseudo-bulk
Label	heart disease	severity
Input	bulk counts, 27,693 genes	pseudo-bulk counts, 148 patients
Pathway mask	Reactome curated (depth 2, ≥ 15)	Reactome immune
$K_{\text{pathway}} + K_{\text{de novo}}$	309 + 300	358 + 50
Auxiliary covariates	5 (sex, BMI, smoking, hypertension, T2D)	sex, 3 comorbidities
α_{θ}	0.6814	2.4135
α_{β}	2.0062	1.1470
α_{ξ}	1.0 (default)	1.0 (default)
α_{η}	1.0 (default)	1.0 (default)
b_v	0.2413	1.5999
σ_{γ}	1.2273	1.4595
α_{π}	0.2407	1.0371
$\beta_{\pi, \text{scale}}$	5.1838	36.8330
Max. CAVI iterations	500	500
Relative tolerance	10^{-3}	10^{-3}
Supervised warm-up	50 iterations	50 iterations
Early stopping	ELBO	ELBO
Regression design	normalized	standard

The programs interpreted in §3.4 were selected by a cross-seed reproducibility filter, applied separately to each cohort.

For interpretation, we use a deliberately larger program budget than prediction: the tuned predictive models allocate only 20 (GTEX) and 50 (COVID-19) free factors, which is enough to maximize held-out AUC but too few to separate distinct biological signals, so we refit each

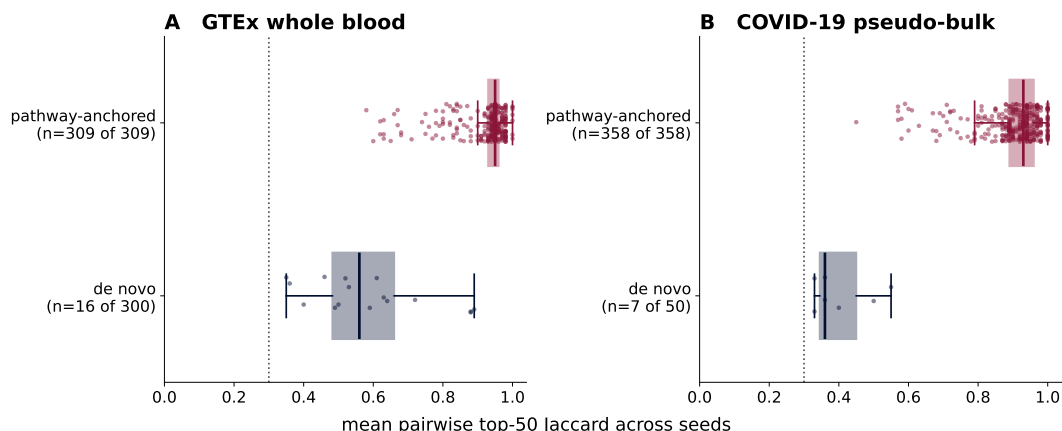


Fig. E6. **Cross-seed stability of the interpreted gene programs.** Mean pairwise Jaccard between the top-50 gene sets of a program’s members across five random seeds, for **(A)** GTEx whole blood (309 pathway-anchored + 300 free factors), and **(B)** COVID-19 pseudo-bulk (358 pathway-anchored + 50 free factors), both in combined mode. Boxes show the median and interquartile range with whiskers to $1.5 \times \text{IQR}$; points are individual programs. The dotted line marks the selection threshold ($J \geq 0.30$, together with presence in ≥ 3 of 5 seeds). Axis labels give the number of programs retained out of the number available in each class.

cohort with a larger free budget and read the programs from those fits. Table E1 gives the exact settings. Note that predictive results are never reported from these runs, and program-level results are never reported from the tuned predictive runs of Table D2.

The two program classes require different matching rules. A pathway factor is basically identified by its name, which is constant across seeds, and only the ranking of genes within the mask can vary. A *de novo* program has no fixed identity and hence, those programs are grouped into cross-seed clusters, where each factor joins an existing cluster with the highest mean Jaccard to its members, provided that the mean is at least 0.30, and otherwise seeds a new cluster. A program cluster is then called reproducible when it is present in at least 3 of the five initialization seeds (Fig.E6).

For each reproducible program we report a consensus top-50 list, ranking genes by their mean $\hat{s}_{g\ell}$ across seeds in which the program appears. Ensembl identifiers that could not be mapped to a gene symbol are reported as identifiers.

All pathway-masked programs pass the filter in both cohorts. This is expected rather than informative: because masked entries are held at $m_{g\ell} = 0$ and excluded from all updates (§B), a masked program’s gene support cannot change between seeds, and the Jaccard statistic measures only the stability of the within-mask ranking. The reproducibility result that carries information is therefore the *de novo* one, and it is far more selective.

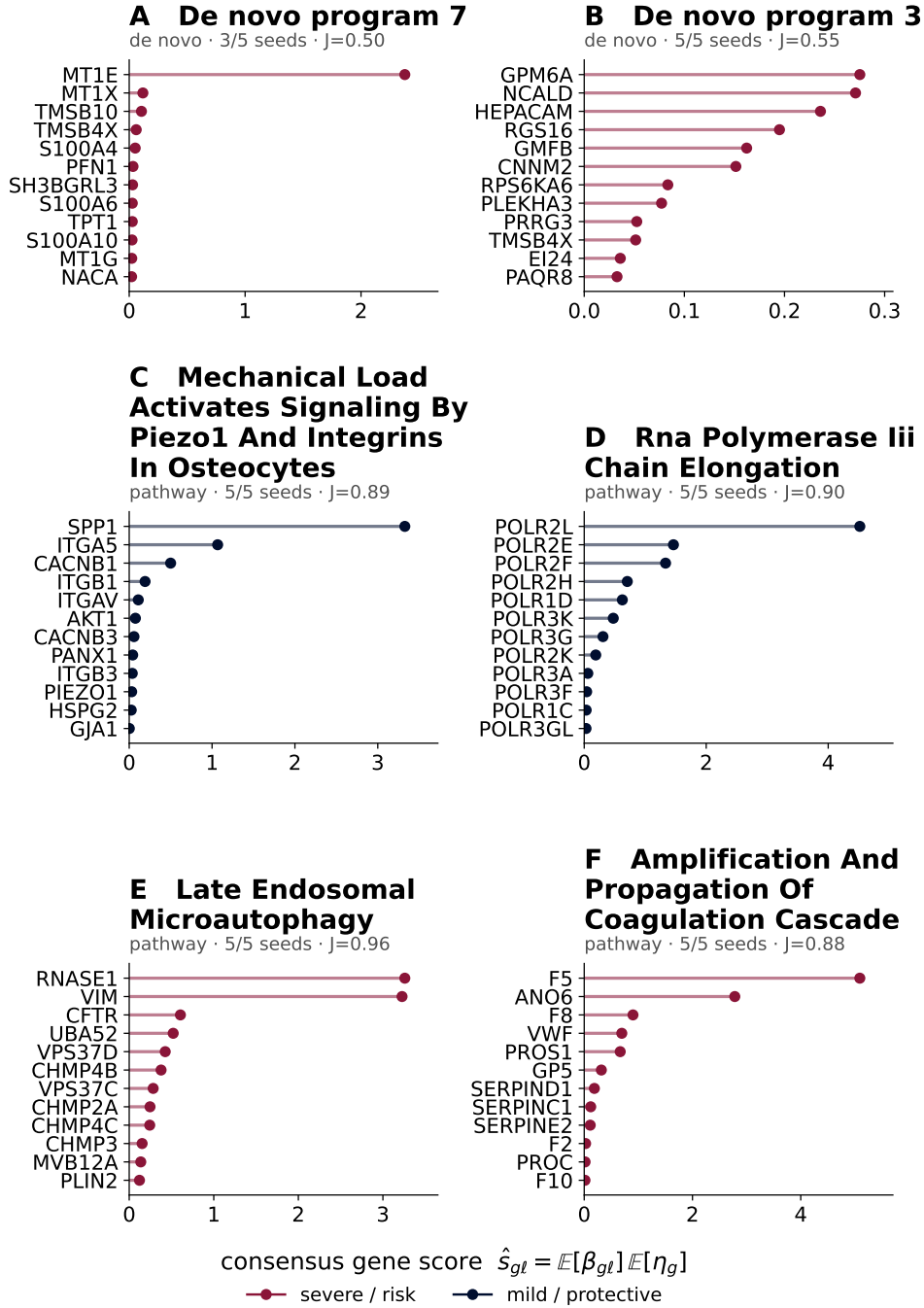


Fig. E7. **Reproducible COVID-19 gene programs and their leading genes.** The six programs with the largest effect-scaled importance $I_\ell = |\bar{v}_{k\ell}| \text{sd}(\bar{\theta}_\ell)$ among those passing the reproducibility filter(Appendix §E.3), from the pseudo-bulk combined fit with the Reactome immune mask. Each panel shows the top 15 genes of the consensus signature, ranked by the mean score $\hat{s}_{gl}$ across the seeds in which the program recurs. Panel headers give the program kind (*de novo* GP or pathway), the number of seeds in which it was found, and its mean pairwise top-50 Jaccard. Colour gives the sign of the supervised coefficient: maroon for programs weighted towards severe disease, navy for programs weighted towards mild disease.

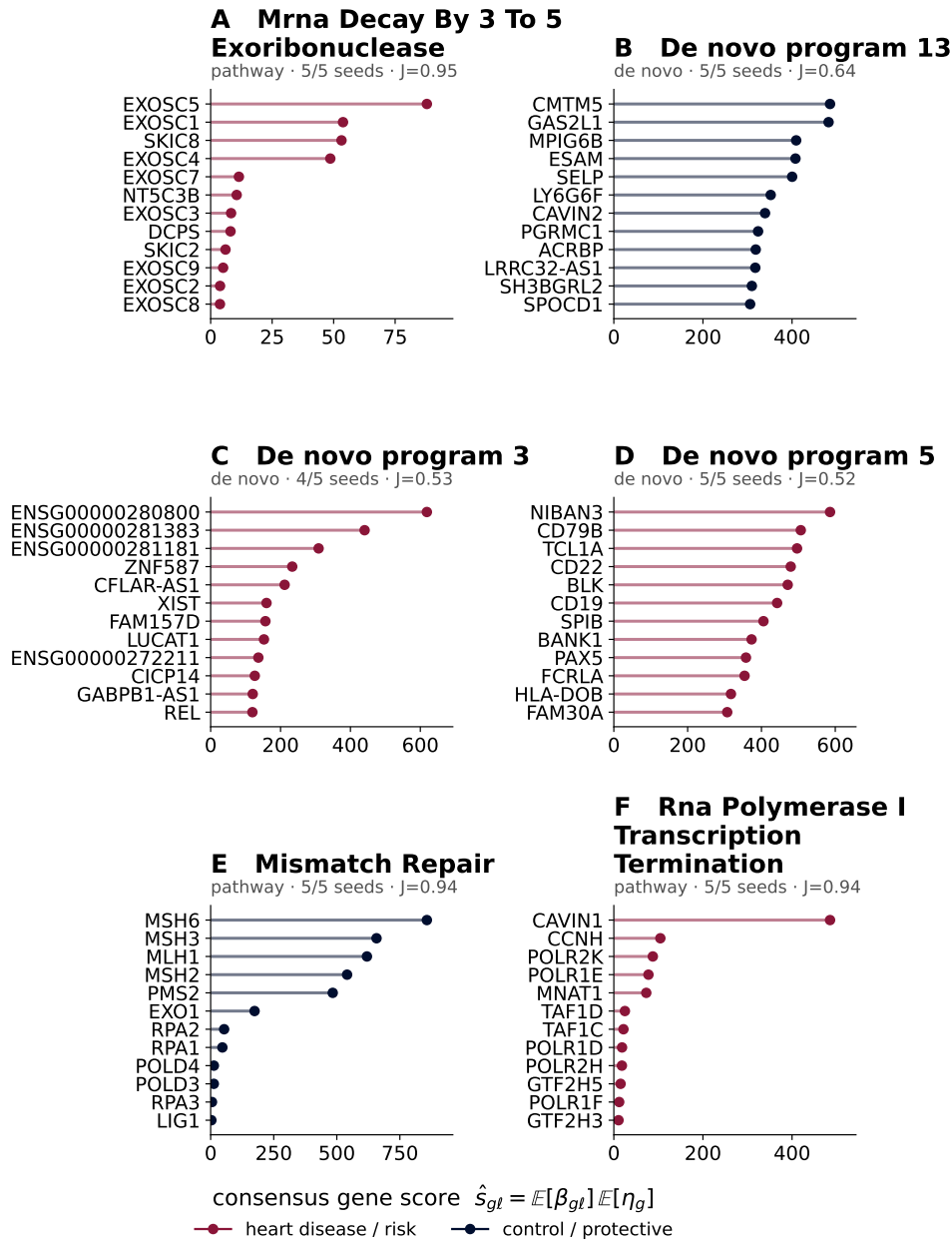


Fig. E8. **Reproducible GTEx heart disease gene programs and their leading genes.** The six programs with the largest effect-scaled importance I_ℓ among those passing the reproducibility filter (Appendix §E.3), from the GTEx heart disease combined fit with the Reactome immune mask. Panel headers give the program kind (*de novo* GP or pathway), the number of seeds in which it was found, and its mean pairwise top-50 Jaccard. Colour gives the sign of the supervised coefficient: maroon for programs weighted towards heart disease cases, navy for programs weighted towards controls.

Table E2. Disease-related genes within disease-associated gene programs.

Program	Dataset	Interpretation	Genes
GP7	COVID-19	Cell proliferation	<i>MKI67, TOP2A, CKS2, ZWINT, CENPF, CEP55, CCNB2, CDKN3, NCAPH, NUSAP1, PKMYT1</i>
GP7	COVID-19	Monocyte and macrophage markers	<i>MT1X, MT1G, S100A4, S100A6, LGALS1, COTL1, SRGN, HMG2</i>
GP7, GP3	COVID-19	Immune-regulatory metallothionein proteins	<i>MT1A, MT1E, MT1F, MT1G, MT1H, MT1M, MT1X</i>
GP7, GP3	COVID-19	Proteins that require binding to zinc or other metals to function	<i>ADNP, ARIH1, CDCA7, MAST4, NCALD, PAQR8, PKMYT1, PPM1N, PSPH, RPS6KA6, S100A4, S100A6, TOP2A, TRIM13, ZBTB20</i>
GP13	GTEEx	Cell adhesion	<i>CLDN5, EGFL7, MMRN1, SELP, TGFB11</i>
GP3	GTEEx	Megakaryocyte or platelet markers	<i>CAVIN2, CMTM5, GP6, PEAR1, SELP</i>
GP5	GTEEx	B cell markers	<i>BANK1, BLK, CD19, CD22, CD72, CD79B, FCRL1, FCRL2, FCRL5, FCRLA, HLA-DOB, IGHD, NIBAN3, PAX5, PLPPR5, TCL1A, VPREB3</i>

Appendix F. Runtime Analysis

We benchmarked wall-clock runtime across all four evaluation settings against the factorization baselines Spectra and scHPF and the non-factorization baselines (Figure F1; five seeds per method). In all experiments DRGP is the fastest factorization method, with median of 17 min on GTEx whole blood. On the bulk and single-cell simulations it is 4–5 $\times$ faster than scHPF and, on the single-cell simulation, more than an order of magnitude faster than Spectra (3 vs. 92 min). These gains follow from DRGP’s GPU implementation: the coordinate-ascent updates are expressed as fused, just-in-time compiled array operations over the sparse count matrix, so runtime scales with the number of nonzero entries rather than the dense cell by gene product. The advantage is most consequential at single-cell scale: on the full COVID-19 cohort (148 patients, $\sim$ 830k cells) DRGP factorizes the data on a single GPU, whereas scHPF exhausts available memory beyond 30 patients on a 500GB node, and Spectra required a 900GB node to complete the factorization on the full cohort, whereas DRGP required 4.7GB of memory on a single GPU.

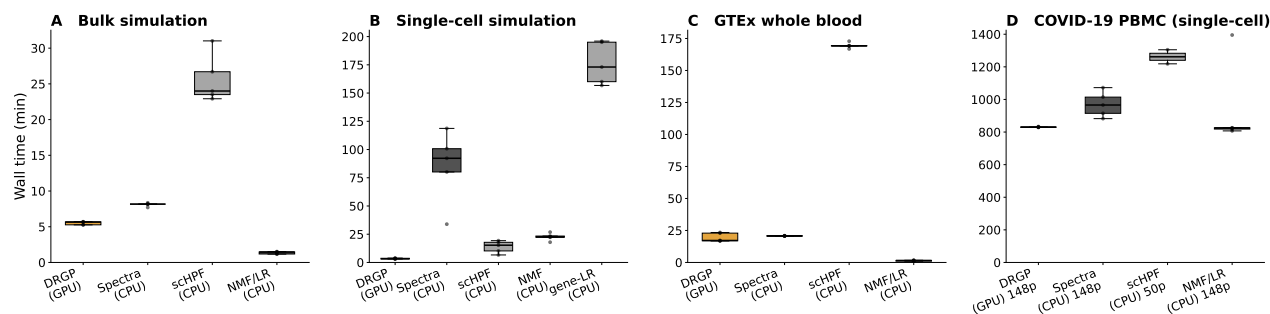


Fig. F1. **Wall-clock runtime of DRGP (combined mode) against the baseline methods** Runtime minutes on **A)** bulk RNA-seq simulations, **B)** single-cell simulations, **C)** GTEx whole blood bulk RNA-seq, and **D)** COVID-19 single-cell RNA-seq. Boxes show the median and interquartile range over five random seeds, whiskers extend to 1.5 $\times$ IQR, and points are the individual seeds. DRGP runs on a single GPU; scHPF, NMF, and the logistic-regression baselines run on CPU. on GTEx, Spectra was run on the 10,000 most-expressed genes while DRGP and the remaining baselines used the full 27,693-gene axis, and on COVID-19 the baselines used 13,770 genes. On COVID-19 PBMC each method is shown at the largest patient cohort it can process without exhausting memory: DRGP and the logistic/NMF baselines complete the full 148-patient cohort ($\sim$ 830k cells), whereas scHPF exhausts memory beyond 30 patients. Simulation and GTEx values are directly comparable across methods.

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